

DONGYEUN LEE

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EDUCATION

Mar. 2026 -	Korea Advanced Institute of Science and Technology (KAIST) PhD. in School of Electrical Engineering Advisor: Prof. Junmo Kim	Daejeon, South Korea
Mar. 2021 - Aug. 2022	Korea Advanced Institute of Science and Technology (KAIST) M.S. in School of Electrical Engineering Advisor: Prof. Junmo Kim	Daejeon, South Korea
Mar. 2016 - Feb. 2021	Korea Advanced Institute of Science and Technology (KAIST) B.S. in School of Computing	Daejeon, South Korea

WORK EXPERIENCE

Apr. 2024 - Feb. 2026	Korea Advanced Institute of Science and Technology (KAIST) Research Scientist (Alternative military service)	Daejeon, South Korea
May 2022 - Mar. 2024	Klleon AI Research AI Researcher (Alternative military service). Worked on a virtual human avatar.	Seoul, South Korea
Jan. 2020 - Jun. 2020	Vision and Learning Laboratory, KAIST Research Intern.	Daejeon, South Korea
Apr. 2019 - Aug. 2019	Dabeeo Research Intern. Worked on Arial Image Change Detection and Object Detection.	Seoul, South Korea

PUBLICATIONS

HyperVAttention: Efficient Sparse Attention with Spatio-Temporal Clustering for Video Diffusion <u>Dongyeun Lee</u> , Amir Zandieh, Vahab Mirrokni, Junmo Kim [†] , Insu Han [†]	Under Review
Mix-up Your Samples: Implicit Discriminator for Enhancing Guidance-Free Visual Generative Models Jiwan Hur, Dong-Jae Lee, <u>Dongyeun Lee</u> , Gyojin Han, Junmo Kim	Under Review
ConceptPrism: Concept Disentanglement in Personalized Diffusion Models via Residual Token Optimization Minseo Kim, Minchan Kwon, <u>Dongyeun Lee</u> , Yunho Jeon, Junmo Kim	CVPR 2026
Inlier-Centric Post-Training Quantization for Object Detection Models Minsu Kim, <u>Dongyeun Lee</u> , Jaemyung Yu, Jiwan Hur, Giseop Kim, Junmo Kim	ICLR 2026
DMQ: Dissecting Outliers of Diffusion Models for Post-Training Quantization <u>Dongyeun Lee</u> , Jiwan Hur, Hyounguk Shon, Jae Young Lee, Junmo Kim	ICCV 2025
Comparison Reveals Commonality: Customized Image Generation through Contrastive Inversion Minseo Kim, Minchan Kwon, <u>Dongyeun Lee</u> , Yunho Jeon, Junmo Kim	CVPRW 2025
RADIO: Reference-Agnostic Dubbing Video Synthesis <u>Dongyeun Lee</u> [*] , Chaewon Kim [*] , Sangjoon Yu, Jaejun Yoo [†] , Gyeong-Moon Park [†]	WACV 2024
Training Cartoonization Networks without Cartoon Doyeon Kim, <u>Dongyeun Lee</u> , Donggyu Joo, Junmo Kim	ICIP 2023
Fix the Noise: Disentangling Source Feature for Controllable Domain Translation <u>Dongyeun Lee</u> , Jae Young Lee, Doyeon Kim, Jaehyun Choi, Jaejun Yoo, Junmo Kim	CVPR 2023
Fix the Noise: Disentangling Source Feature for Transfer Learning of StyleGAN <u>Dongyeun Lee</u> , Jae Young Lee, Doyeon Kim, Jaehyun Choi, Junmo Kim	CVPRW 2022 (<i>Best Paper Award</i>)

SELECTED PROJECTS

Jan. 2024 - Mar. 2024	GPU-Accelerated Data Pipeline for Lip-Sync Discriminator Optimized CPU-bound training pipeline of a lightweight model with heavy data processing and loading. Developed a high-throughput data pipeline using NVIDIA DALI, enabling batched GPU-side image/audio preprocessing and direct GPU loading. Achieved up to 28.63x training speedup. Tech: Python, PyTorch, NVIDIA DALI	Klleon AI Research
May 2022 - Mar. 2024	High-Fidelity Audio Driven Talking Head Synthesis Led end-to-end development of core lip-sync pipeline including data collection, pre-processing, model architecture design, training, training pipeline optimization, and deployment. Improved output resolution by 2x and enhanced image fidelity by 13% over the prior internal baseline. Tech: Python, Pytorch. Applied product: Chat-Avatar, Klone Studio	Klleon AI Research

Apr. 2019 - Aug. 2019

Arial Image Change Detection and Object Detection

Dabeoo

Developed a change and object detection system for illegal structure detection and monitoring urban changes in Seoul and Daejeon. Overcame challenges in small-object detection and noisy backgrounds through iterative model optimization, improving F1 score from **0.60** to **0.95**.

Tech: Python, Keras.

SKILLS

ML Algorithm

Python, Pytorch, Pytorch3D, Keras, Tensorflow

ML Efficiency

Triton, C++, CUDA, NVIDIA DALI, Nsight

Miscellaneous

Docker, Bash

Basic Knowledge

R, Node.js, JavaScript, MongoDB, MySQL, Scala, Kotlin

SOFTWARE ARTIFACTS

FixNoise

 [LeeDongYeun/FixNoise](#)

Pytorch implementation of FixNoise.

Deepfake Detection Framework

 [LeeDongYeun/deepfake-detection](#)

Basic model training and datasets processing code for Deepfake Detection.

Keras M2Det

 [LeeDongYeun/keras-m2det](#)

Keras implementation of *M2Det: A Single-Shot Object Detector based on Multi-Level Feature Pyramid Network*.

HONERS AND AWARDS

Jun. 2022

Best Paper Award

Sponsored by Google

CVPR Workshops on AI for Content Creation.